

9(a)	difference between mass of nucleus and (total) mass of nucleons	M1
	when infinitely separated	A1
9(b)(i)	neutron	B1
9(b)(ii)	$E = \Delta m c^2$	C1
	$\Delta m = (0.030377 - 0.002388 - 0.009105)u$ (= 0.018884u)	C1
	energy release = $(0.030377 - 0.002388 - 0.009105) \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 = 2.8 \times 10^{-12} \text{ J}$	A1
9(c)(i)	number of atoms per unit time = $(1.4 \times 10^{28}) / (2.8 \times 10^{-12})$ (= $5.0 \times 10^{39} \text{ s}^{-1}$)	C1
	mass of one atom = $4 \times 1.66 \times 10^{-27}$ or $(4 \times 10^{-3}) / (6.02 \times 10^{23})$ (= $6.64 \times 10^{-27} \text{ kg}$)	C1
	mass per unit time = $6.64 \times 10^{-27} \times 5.0 \times 10^{39}$ = $3.3 \times 10^{13} \text{ kg s}^{-1}$	A1
9(c)(ii)	$L = 4\pi\sigma r^2 T^4$	C1
	$1.4 \times 10^{28} = 4\pi \times 5.67 \times 10^{-8} \times (2.3 \times 10^9)^2 \times T^4$	
	$T = 7800 \text{ K}$	A1